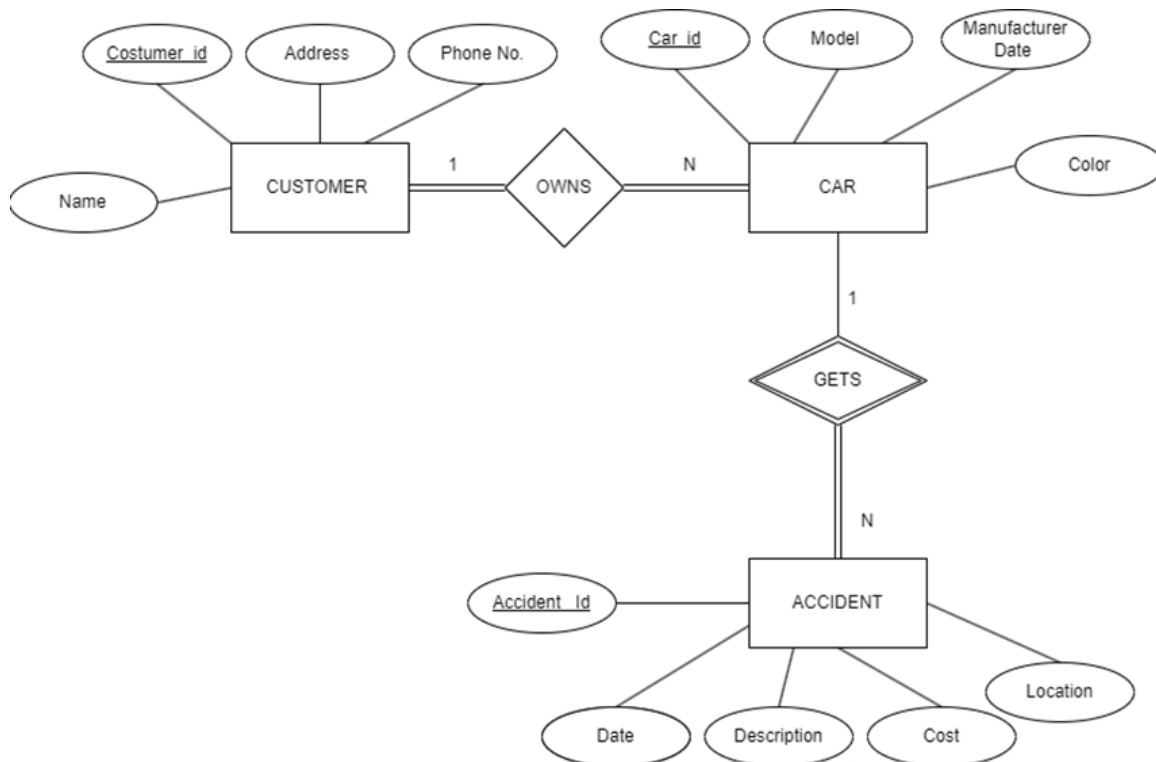


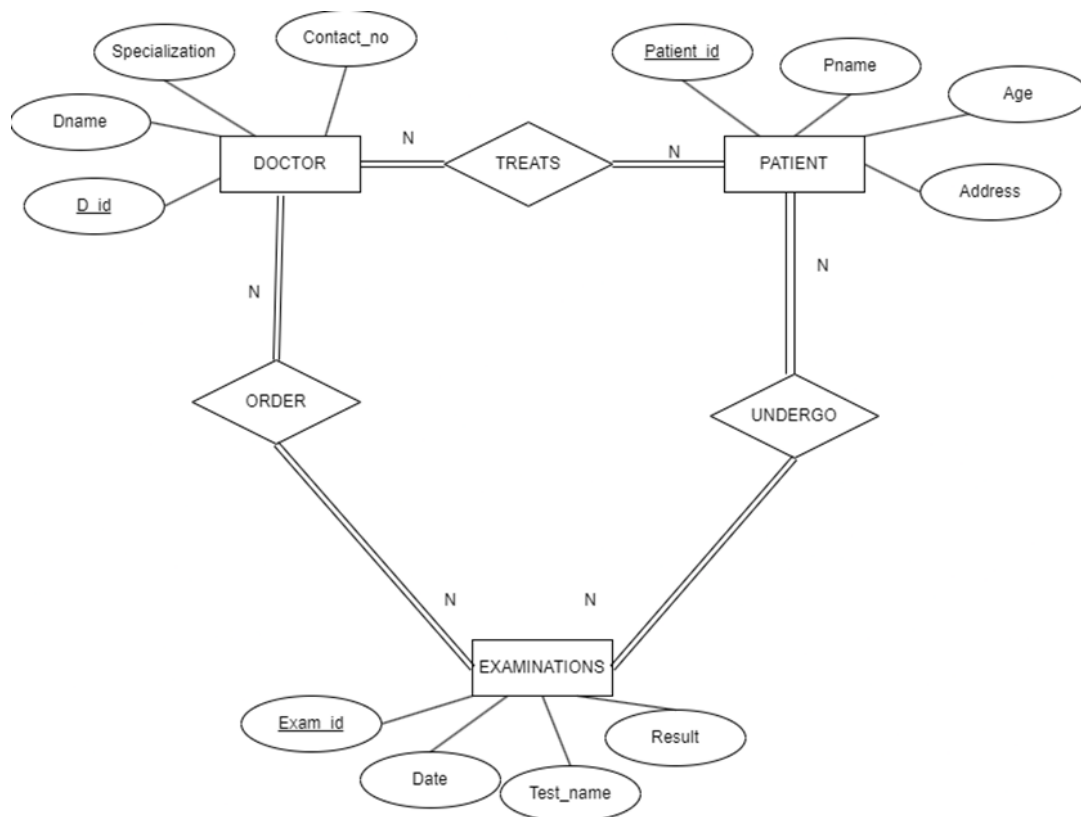
S. N	Name of Practical and Assignment	Signature
1.	Construct an ER diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.	
2.	Construct an ER diagram for a hospital with a set of patients and a set of doctors. Associate with each patient a log of the various tests and examinations conducted.	
3.	Construct an ER diagram of the library system at your college.	
4.	Construct an ER diagram to maintain data about students, instructors, semester, and courses in a college.	
5.	Design an ER schema for keeping track of information about marks that students get in different exams of different courses.	
6.	A database is being constructed to keep track of the teams and games of a sports league. A team has a number of players, not all of whom participate in each game. It is desired to keep track of the players participating in each game for each team, the positions they played in that game, and the result of the game. Design an ER schema for this application.	
7.	Design an ER schema for the bank. The bank has multiple branches and, in each branch, customers open account and borrow loans.	
8.	Consider the university database for maintaining information concerning students, teachers, courses, and grades in a university environment as given below. TEACHER(TID, Tname, Taddress, Salary, Qualification) TEACHES(TID, SID) STUDENT(SID, Sname, Saddress, Section) GRADE_REPORT(SID, CID, Grade) COURSE(CID, Cname, Credit_hours)	

Lab 1: Designing ER Diagrams

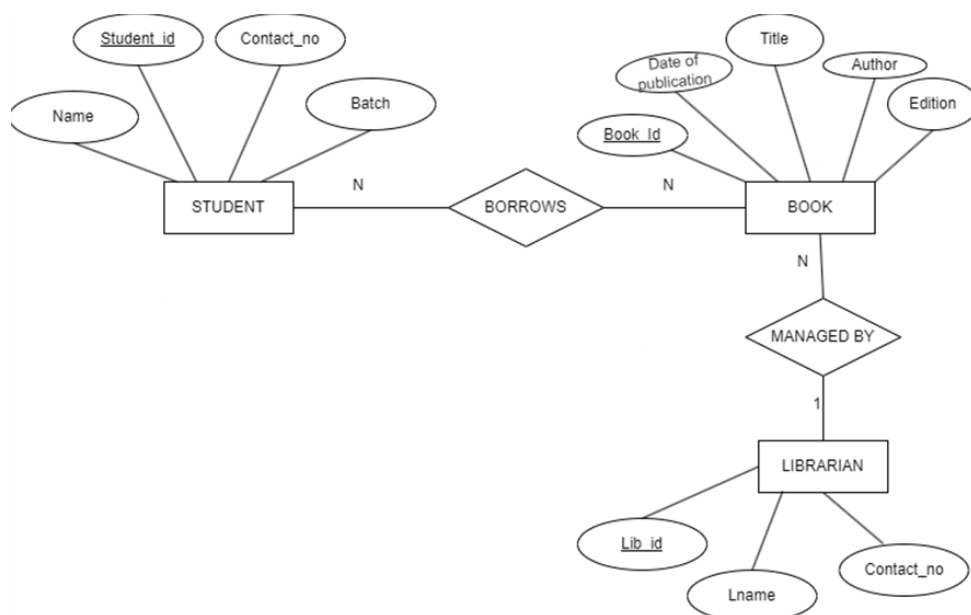
1. Construct an ER diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.



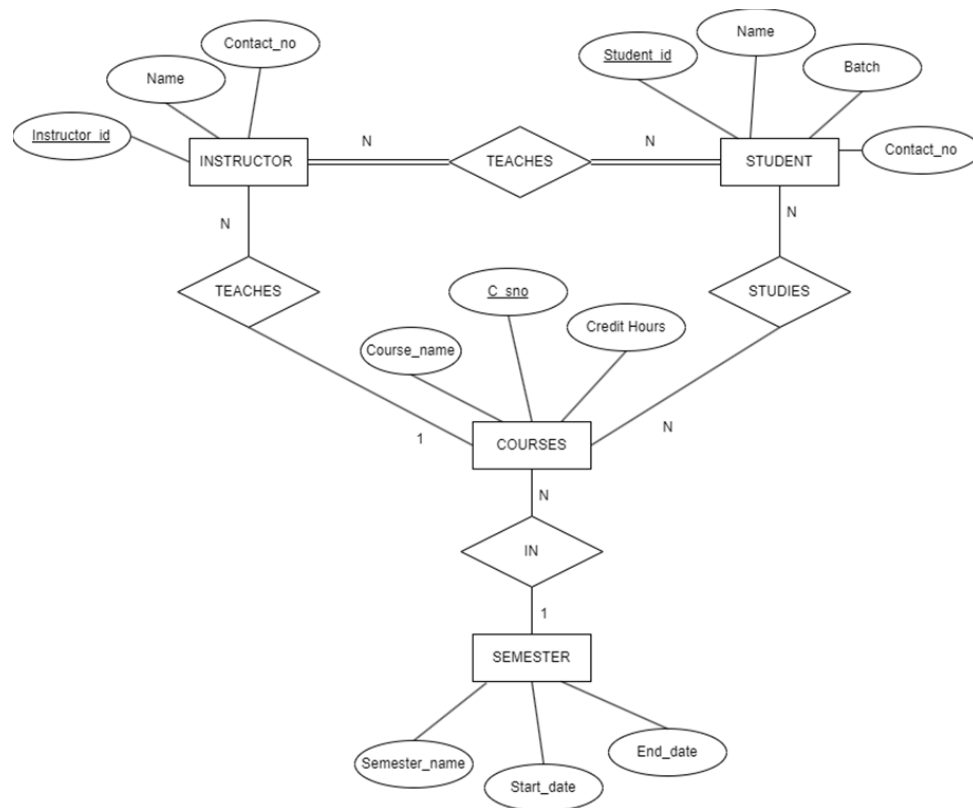
2. Construct an ER diagram for a hospital with a set of patients and a set of doctors. Associate with each patient a log of the various tests and examinations conducted.



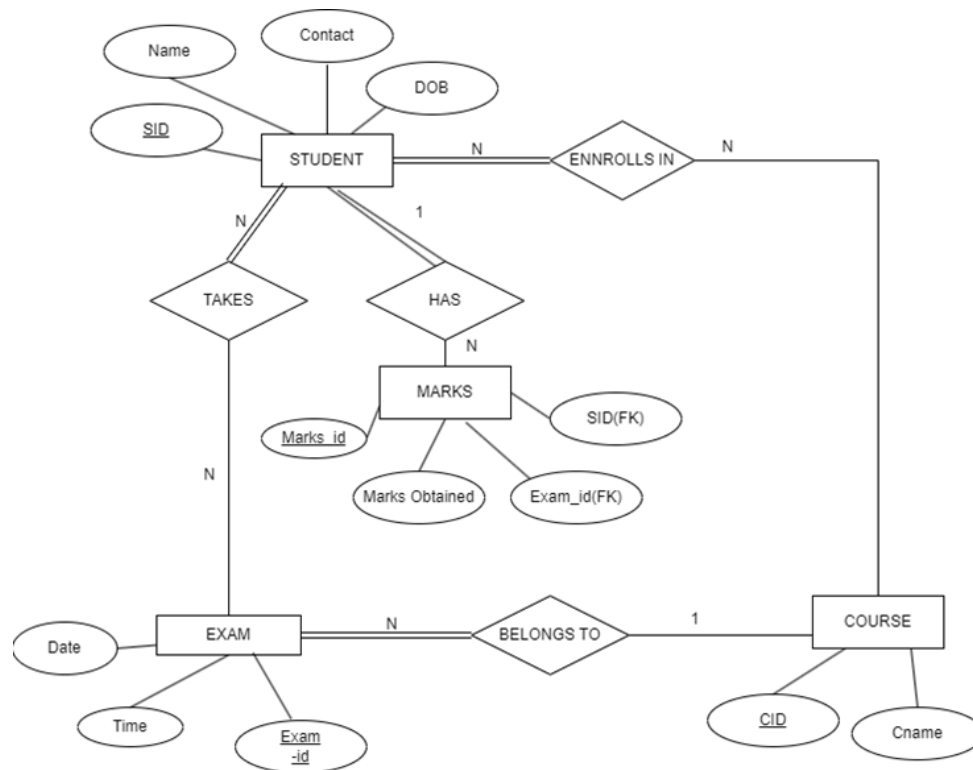
3. Construct an ER diagram of the library system at your college.



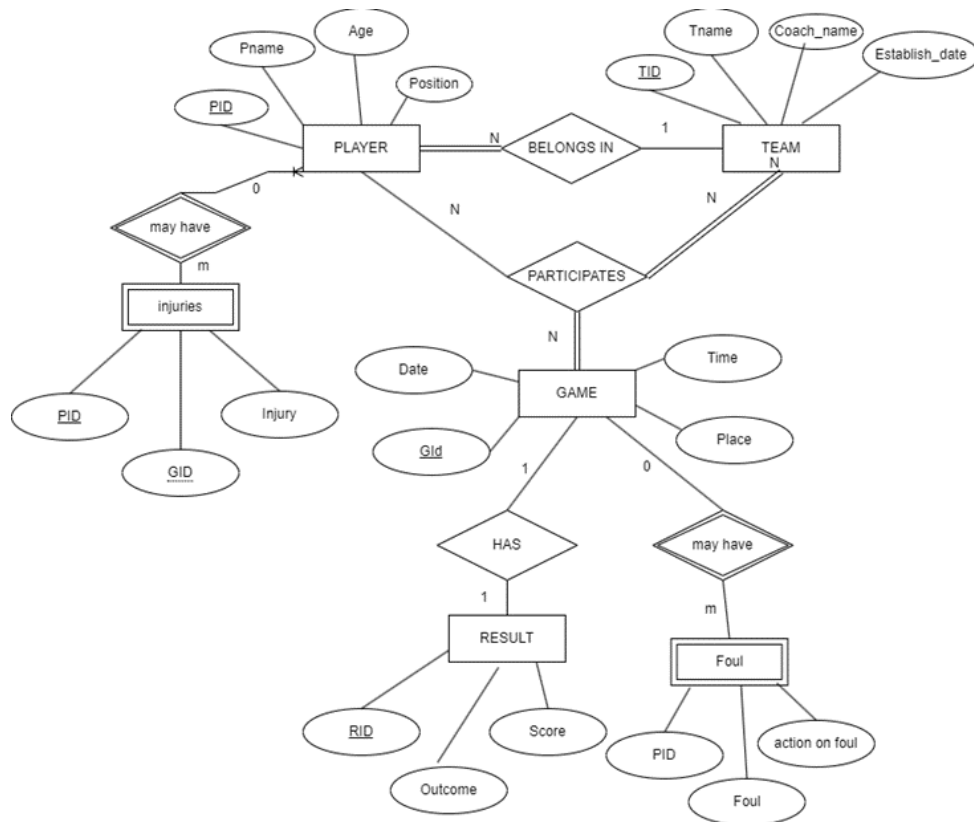
4. Construct an ER diagram to maintain data about students, instructors, semester, and courses in a college.



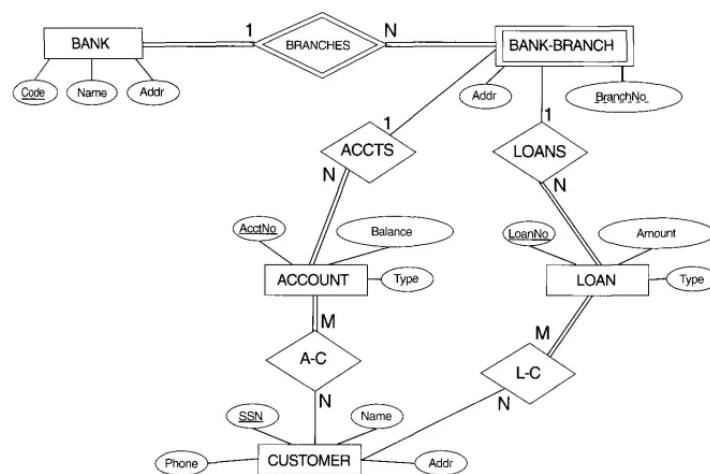
5. Design an ER schema for keeping track of information about marks that students get in different exams of different courses.



6. A database is being constructed to keep track of the teams and games of a sports league. A team has a number of players, not all of whom participate in each game. It is desired to keep track of the players participating in each game for each team, the positions they played in that game, and the result of the game. Design an ER schema for this application.



7. Design an ER schema for the bank. The bank has multiple branches and, in each branch, customers open account and borrow loans.



BANK Database ER Schema

BANK

Attribute	Type	Constraint
Code	INT	Primary Key
Name	VARCHAR(50)	
Addr	VARCHAR(100)	

BANK_BRANCH

Attribute	Type	Constraint
BranchNo	INT	Primary Key
Addr	VARCHAR(100)	
Bank_Code	INT	Foreign Key (BANK.Code)

ACCOUNT

Attribute	Type	Constraint
AcctNo	INT	Primary Key
Balance	DECIMAL(10,2)	
Type	VARCHAR(20)	
BranchNo	INT	Foreign Key (BANK_BRANCH.BranchNo)

LOAN

Attribute	Type	Constraint
LoanNo	INT	Primary Key
Amount	DECIMAL(10,2)	
Type	VARCHAR(20)	
BranchNo	INT	Foreign Key (BANK_BRANCH.BranchNo)

CUSTOMER

Attribute	Type	Constraint
SSN	VARCHAR(9)	Primary Key
Name	VARCHAR(50)	
Phone	VARCHAR(15)	
Addr	VARCHAR(100)	

ACCOUNT_CUSTOMER

Attribute	Type	Constraint
AcctNo	INT	Foreign Key (ACCOUNT.AcctNo)
SSN	VARCHAR(9)	Foreign Key (CUSTOMER.SSN)

LOAN_CUSTOMER

Attribute	Type	Constraint
LoanNo	INT	Foreign Key (LOAN.LoanNo)

SSN	VARCHAR(9)	Foreign Key (CUSTOMER.SSN)
-----	------------	----------------------------

Lab 2: University Database

Consider the university database for maintaining information concerning students, teachers, courses, and grades in a university environment as given below.

```
TEACHER(TID, Tname, Taddress, Salary, Qualification)
TEACHES(TID, SID)
STUDENT(SID, Sname, Saddress, Section)
GRADE_REPORT(SID, CID, Grade)
COURSE(CID, Cname, Credit_hours)
```

a. Create all the tables in the database with proper primary key, foreign key, and referential integrity constraints.

```
CREATE DATABASE UNIVERSITY;
USE UNIVERSITY;

CREATE TABLE TEACHER (
    TID INT PRIMARY KEY,
    Tname VARCHAR(100) NOT NULL,
    Taddress VARCHAR(200) NOT NULL,
    Salary DECIMAL(10,2) NOT NULL CHECK (Salary >0),
    Qualification VARCHAR(100) NOT NULL
);

CREATE TABLE STUDENT (
    SID INT PRIMARY KEY,
    Sname VARCHAR(100) NOT NULL,
    Saddress VARCHAR(200) NOT NULL,
    Section VARCHAR(50) NOT NULL
);

CREATE TABLE COURSE (
    CID INT PRIMARY KEY,
    Cname VARCHAR(100) NOT NULL UNIQUE,
    Credit_hours INT NOT NULL CHECK (Credit_hours > 0)
```

```

);

CREATE TABLE TEACHES (
    TID INT,
    SID INT,
    PRIMARY KEY (TID, SID),
    FOREIGN KEY (TID) REFERENCES TEACHER(TID) ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (SID) REFERENCES STUDENT(SID) ON DELETE CASCADE ON UPDATE CASCADE
);

CREATE TABLE GRADE_REPORT (
    SID INT,
    CID INT,
    Grade CHAR(2) NOT NULL CHECK (Grade IN ('A', 'A-', 'B+', 'B', 'B-', 'C+', 'C', 'C-', 'D+', 'D', 'D-', 'F+', 'F', 'F-', 'P', 'NP', 'S')),
    PRIMARY KEY (SID, CID),
    FOREIGN KEY (SID) REFERENCES STUDENT(SID) ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (CID) REFERENCES COURSE(CID) ON DELETE CASCADE ON UPDATE CASCADE
);

```

Table: teacher

Columns:

<u>TID</u>	int PK
Tname	varchar(100)
Taddress	varchar(200)
Salary	decimal(10,2)
Qualification	varchar(100)

Table: student

Columns:

<u>SID</u>	int PK
Sname	varchar(100)
Saddress	varchar(200)
Section	varchar(50)

Table: course

Columns:

<u>CID</u>	int PK
Cname	varchar(100)
Credit_hours	int

Table: teaches

Columns:

<u>TID</u>	int PK
<u>SID</u>	int PK

Table: grade_report

Columns:

<u>SID</u>	int PK
<u>CID</u>	int PK
Grade	char(2)

b. Insert appropriate data in the database.

```
INSERT INTO TEACHER (TID, Tname, Taddress, Salary, Qualification) VALUES
(1, 'Nawaraj Paudel', 'Kathmandu', 60000.00, 'PhD in Database Management'),
(2, 'Bikash Balami', 'Pokhara', 55000.00, 'Masters in Information Systems'),
(3, 'Cristiano Adhikari', 'Kathmandu', 52000.00, 'Masters in Computer Science'),
(4, 'Anand KC', 'Lalitpur', 48000.00, 'Masters in CSIT');
```

```
INSERT INTO STUDENT (SID, Sname, Saddress, Section) VALUES
(101, 'Ram Timalina', 'Bhaktapur', 'A'),
(102, 'Nitesh Giri', 'Lalitpur', 'B'),
(103, 'Hari Krishna Bhushal', 'Kathmandu', 'A'),
(104, 'Sandeep Lamichhane', 'Kathmandu', 'B'),
(105, 'Messi Nepali', 'Pokhara', 'A');
```

```
INSERT INTO COURSE (CID, Cname, Credit_hours) VALUES
(201, 'Database Management Systems', 3),
(202, 'Operating System', 4),
(203, 'Theory of computation', 3),
(204, 'Artificial Intelligence', 4),
(205, 'Computer Networks', 3);
```

```
INSERT INTO TEACHES (TID, SID) VALUES
(1, 101), (1, 102), (1, 103),
(2, 103), (2, 104),
(3, 101), (3, 105),
(4, 102), (4, 104), (4, 105);
```

```
INSERT INTO GRADE_REPORT (SID, CID, Grade) VALUES
(101, 201, 'A'), (101, 202, 'B+'), (101, 203, 'A-'), (101, 204, 'B'), (101, 205, 'A'),
(102, 201, 'B'), (102, 202, 'B-'), (102, 203, 'A-'), (102, 204, 'B'), (102, 205, 'B'),
(103, 201, 'A-'), (103, 202, 'B+'), (103, 203, 'B'), (103, 204, 'B'), (103, 205, 'A'),
(104, 201, 'B'), (104, 202, 'B+'), (104, 203, 'B-'), (104, 204, 'A-'), (104, 205, 'B'),
(105, 201, 'C'), (105, 202, 'C+'), (105, 203, 'B'), (105, 204, 'A-'), (105, 205, 'C');
```

```
SELECT * FROM TEACHER;
SELECT * FROM STUDENT;
SELECT * FROM COURSE;
SELECT * FROM TEACHES;
SELECT * FROM GRADE_REPORT;
```

	TID	Tname	Taddress	Salary	Qualification
▶	1	Nawaraj Paudel	Kathmandu	60000.00	PhD in Database Management
	2	Bikash Balami	Pokhara	55000.00	Masters in Information Systems
	3	Cristiano Adhikari	Kathmandu	52000.00	Masters in Computer Science
	4	Anand KC	Lalitpur	48000.00	Masters in CSIT
*	NULL	NULL	NULL	NULL	NULL

TEACHER 11 ×

	SID	Sname	Saddress	Section
▶	101	Ram Timalisina	Bhaktapur	A
	102	Nitesh Giri	Lalitpur	B
	103	Hari Krishna Bhushal	Kathmandu	A
	104	Sandeep Lamichhane	Kathmandu	B
	105	Lionel Messi	Pokhara	A
*	NULL	NULL	NULL	NULL

STUDENT 12 ×

	CID	Cname	Credit_hours
▶	201	Database Management Systems	3
	202	Operating System	4
	203	Theory of computation	3
	204	Artificial Intelligence	4
	205	Computer Networks	3
*	NULL	NULL	NULL

COURSE 13 ×

	SID	CID	Grade
▶	101	201	A
	101	202	B+
	102	201	B
	102	203	A-
	103	201	A-
	103	204	B
	104	202	B+
	104	205	A
	105	203	B
	105	204	A-
*	NULL	NULL	NULL

GRADE_REPORT 10 ×

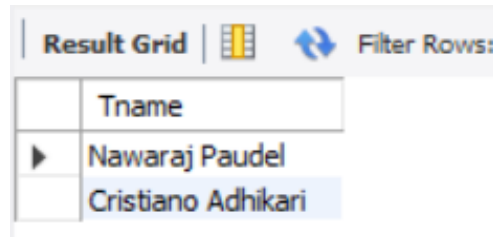
	TID	SID
▶	1	101
	3	101
	1	102
	4	102
	1	103
	2	103
	2	104
	4	104
	3	105
	4	105
*	NULL	NULL

TEACHES 9 ×

c. Retrieve the names of all teachers whose address is Kathmandu. (SQL + Relational Algebra)

```
SELECT Tname FROM TEACHER WHERE Taddress = 'KATHMANDU';
```

```
 $\pi(\text{Tname})(\sigma(\text{Taddress} = \text{'Kathmandu'})(\text{TEACHER}))$ 
```



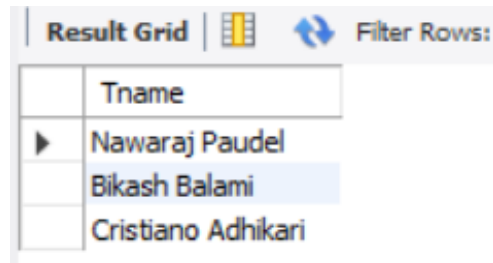
The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. The grid contains a single column titled 'Tname' with two rows: 'Nawaraj Paudel' and 'Cristiano Adhikari'. The second row is highlighted in blue.

	Tname
▶	Nawaraj Paudel
	Cristiano Adhikari

d. Retrieve the names of all teachers whose salary is greater than 50000. (SQL + Relational Algebra)

```
SELECT Tname FROM TEACHER WHERE Salary > 50000;
```

```
 $\pi(\text{Tname})(\sigma(\text{Salary} > 50000)(\text{TEACHER}))$ 
```



The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. The grid contains a single column titled 'Tname' with three rows: 'Nawaraj Paudel', 'Bikash Balami', and 'Cristiano Adhikari'. The second row is highlighted in blue.

	Tname
▶	Nawaraj Paudel
	Bikash Balami
	Cristiano Adhikari

e. Retrieve the names of all students taught by Nawaraj Paudel. (SQL + Relational Algebra)

```
SELECT S.Sname FROM STUDENT S
join TEACHES T ON T.SID = S.SID
join TEACHER TE ON TE.TID = T.TID
WHERE TE.Tname = "Nawaraj Paudel";
```

```
 $\pi(\text{Sname})(\sigma(\text{Tname} = \text{'Nawaraj Paudel'})(\text{STUDENT} \bowtie \text{TEACHES} \bowtie \text{TEACHER}))$ 
```

Result Grid			Filter Rows:
	Sname		
▶	Ram Timalcina		
	Nitesh Giri		
	Hari Krishna Bhushal		

f. Retrieve the names and grades obtained on all courses by the student Ram Timalcina.(SQL + Relational Algebra)

```
select S.Sname, G.Grade, C.Cname From Student S
join grade_report G on G.SID = S.SID
join course C on C.CID = G.CID
Where Sname = "Ram Timalcina";
```

$$\pi(\text{Cname}, \text{Grade})(\sigma(\text{Sname} = \text{'Ram Timalcina'})(\text{STUDENT} \bowtie \text{GRADE_REPORT} \bowtie \text{COURSE}))$$

Result Grid		Filter Rows:	Export:
	Sname	Grade	Cname
▶	Ram Timalcina	A	Database Management Systems
	Ram Timalcina	B+	Operating System
	Ram Timalcina	A-	Theory of computation
	Ram Timalcina	B	Artificial Intelligence
	Ram Timalcina	B+	Computer Networks

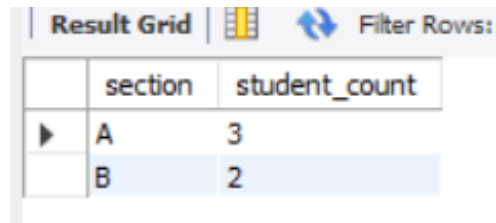
g. Retrieve the average salary of teachers.

```
select avg(salary) as AVG_SALARY from teacher;
```

Result Grid				Filter Rows:
	AVG_SALARY			
	53750.000000			

h. Count the number of students in each section.

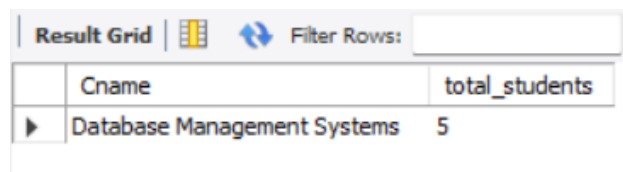
```
select section, count(*) as student_count from student group by Section;
```



	section	student_count
▶	A	3
	B	2

i. Count the number of students who studies DBMS course.

```
select C.Cname, count(S.Sname) as total_students from course C
join grade_report G on C.CID = G.CID
join student S on G.SID = S.SID
where C.cname = "Database Management Systems";
```



	Cname	total_students
▶	Database Management Systems	5